DATA HANDLING AND VISUALIZATION

TABLEAU PRACTICAL ASSESSMENT

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Introduction

The purpose of this assessment was to explore a dataset in Tableau and present the information using charts and a dashboard. The file D_38.csv was used for the practical work. The dataset contains fields such as Month, Region, Sales, Profit, Customers and Discount_pct.

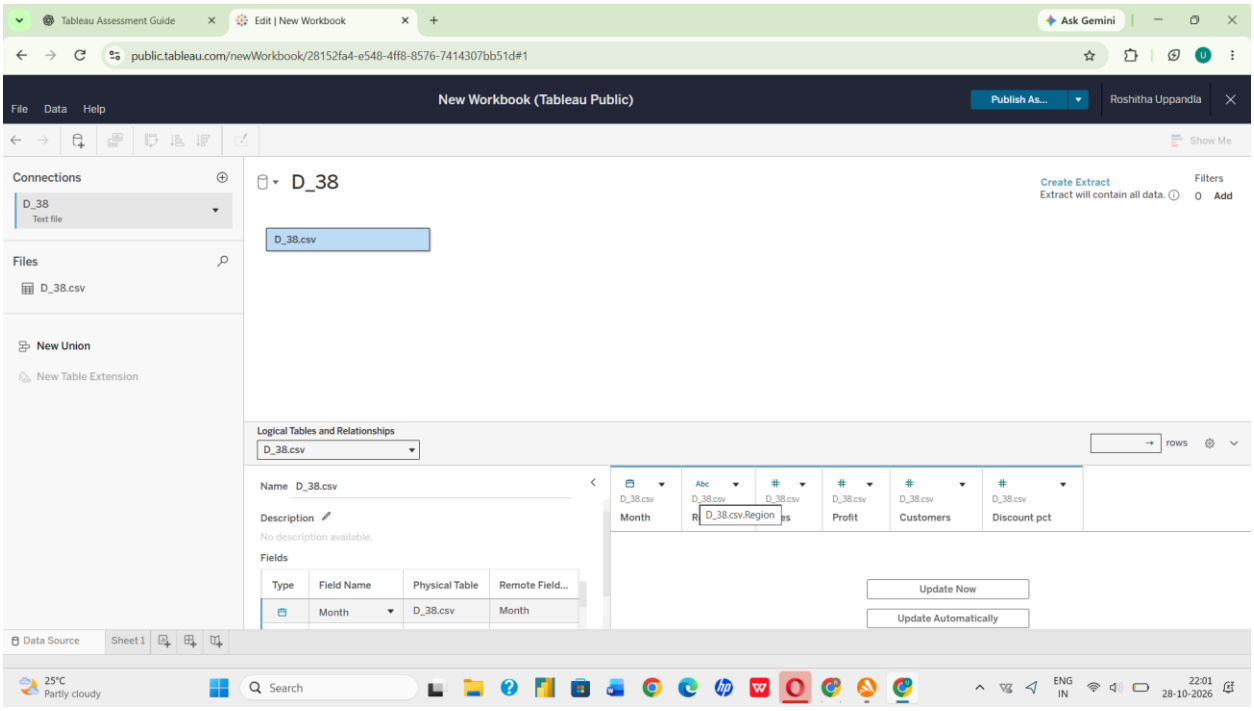
The tasks included connecting the data, creating different visualizations, checking trends, preparing a dashboard and comparing Tableau with R. The charts help to understand the sales and profit information more clearly.

T1: Connect to Data and Check the Dataset

I opened Tableau Public and connected the D_38.csv file using the data connection option. After the file was loaded, I checked the available fields and their data types. Month and Region were used as dimensions, while Sales, Profit, Customers and Discount_pct were used as measures.

I also checked the data in the worksheet before creating the visualizations. This helped me understand which fields could be used for comparing regions and studying monthly sales.

Result: The CSV file was connected successfully and the required fields were available in Tableau.



T2: Bar Chart – Sales by Region

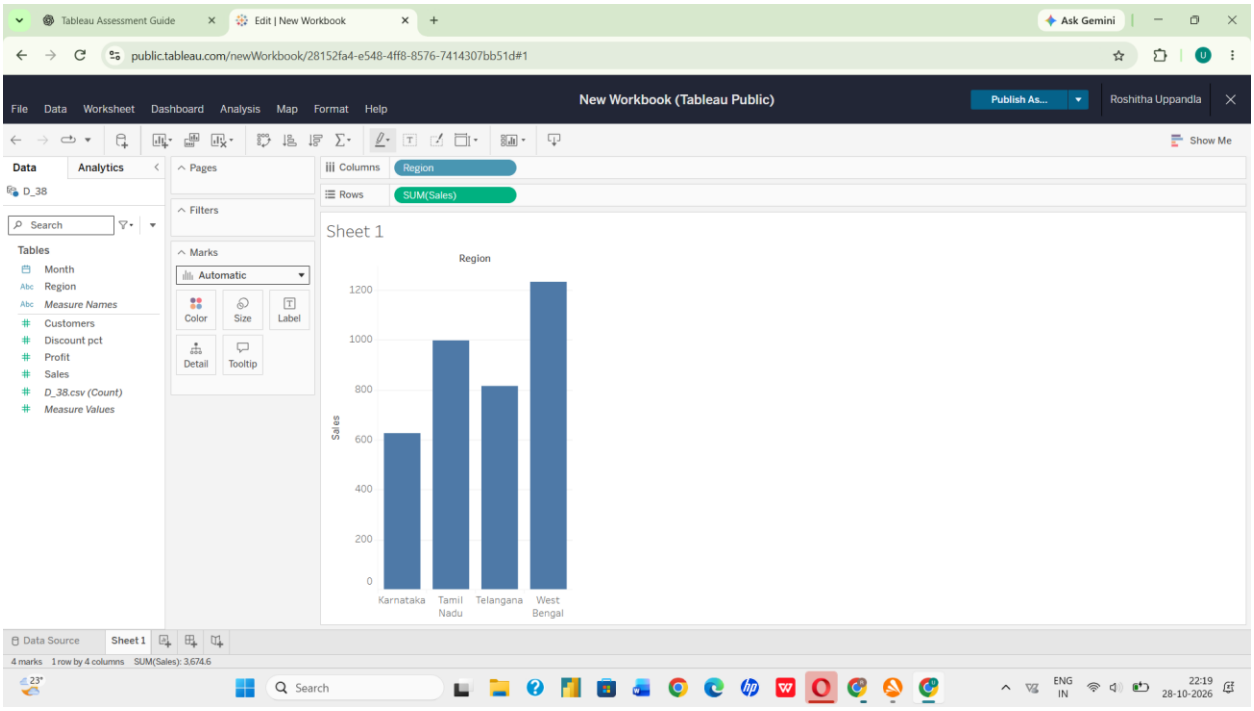
I created a bar chart to compare total sales across the regions. Region was placed on Columns and Sales was placed on Rows. I displayed the sales values on the bars and sorted the regions from highest to lowest so that the comparison would be easy to read.

The chart showed the following sales values:

- West Bengal: 1233.0
- Tamil Nadu: 998.9
- Telangana: 815.7
- Karnataka: 627.0

The total sales shown were 3674.6. West Bengal had the highest sales, while Karnataka had the lowest sales among the four regions. West Bengal contributed approximately 33.56% of the total sales.

Result: The bar chart made it easy to compare the sales of each region and identify the highest and lowest values.



T3: Monthly Sales Trend

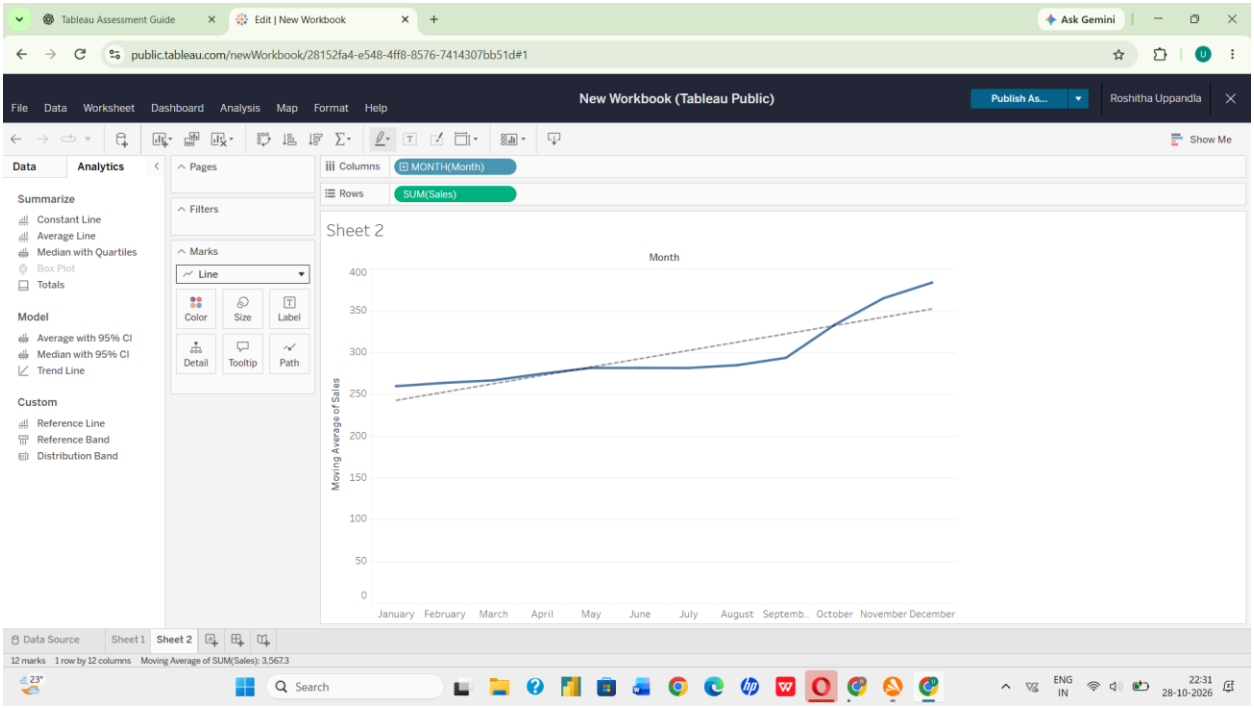
I created a line chart to study how sales changed over the months. Month was placed on Columns and Sales was placed on Rows. I used a moving average to make the general pattern easier to see and added a linear trend line to observe the overall direction of sales.

The line chart is useful because it shows changes over time more clearly than a table of values. The moving average smooths some of the short-term changes, while the trend line gives a general indication of whether sales are increasing or decreasing.

Observation: The exact peak month, its sales value, the June moving-average value and the trend-line slope should be read from the tooltips or trend model in the completed Tableau worksheet.

Result: A monthly sales chart was prepared with a moving average and a linear trend line.

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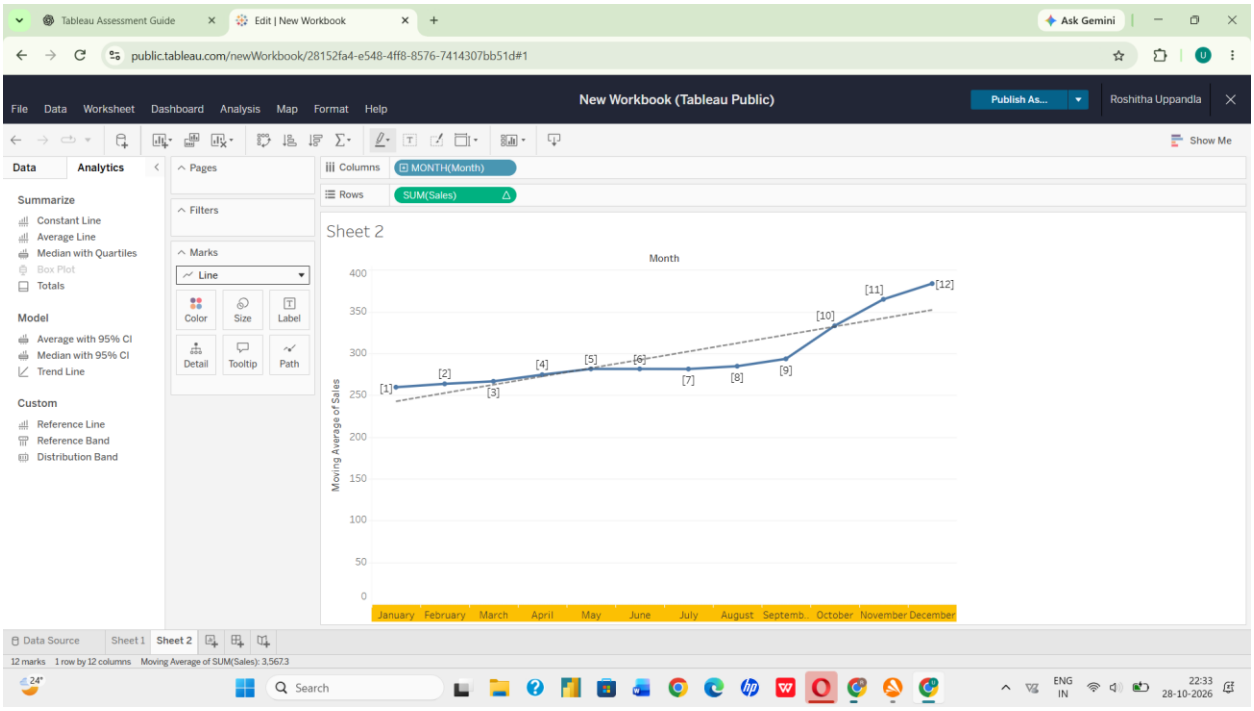


T4: Scatter Plot – Discount and Profit

I prepared a scatter plot to examine the relationship between discount percentage and profit. Discount_pct was placed on the horizontal axis and Profit on the vertical axis. Region was used to distinguish the points by colour, and Month was added to the detail information. A trend line was added to help identify the general relationship between the two measures.

Each point represents a data record. The position of the point shows its discount and profit values. The trend line helps to see whether profit tends to rise or fall as the discount changes. The slope and correlation information should be taken from the trend model displayed in Tableau.

Result: The scatter plot was used to explore the relationship between discount and profit.



T5: Filled Map – Profit by Region

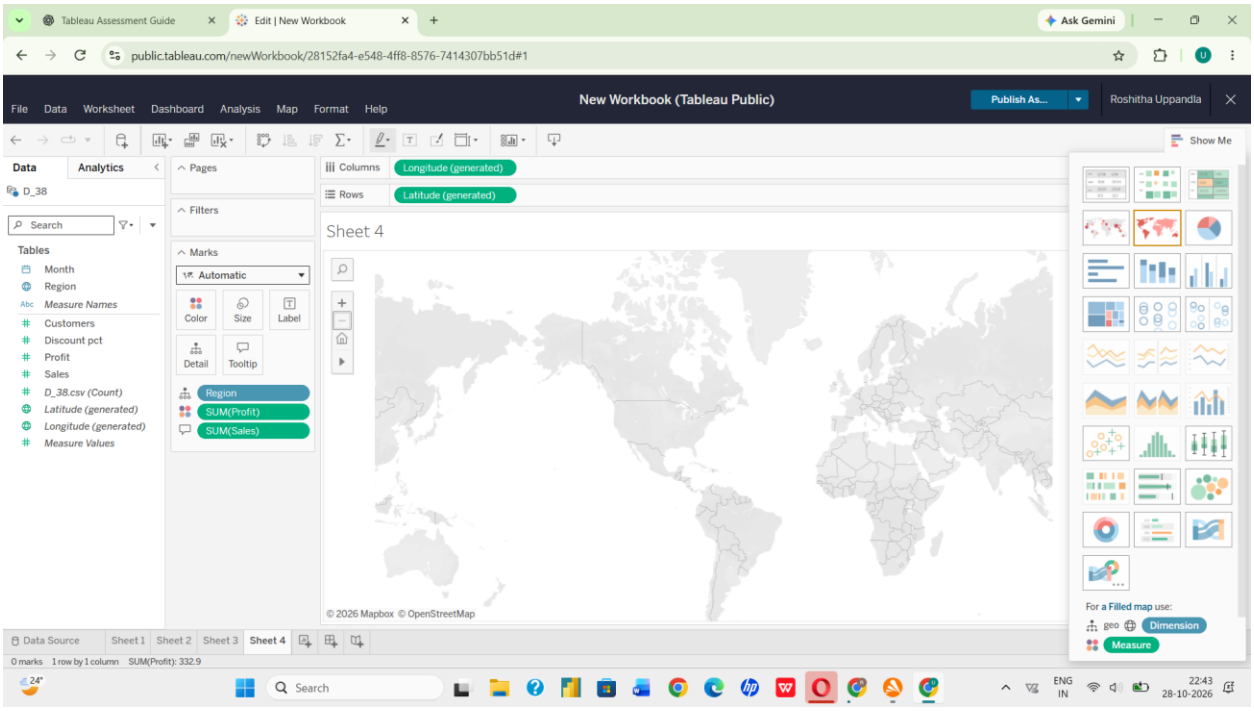
I attempted to show the profit information on a map by using Region as the geographic field and Profit as the measure represented by colour. Sales can also be included in the tooltip so that the sales value appears when a region is selected or hovered over.

The map can help compare the regional profit values geographically. The region with the lowest profit should be identified from the completed map or the underlying data. The overall profit margin can be calculated using the formula:

Profit Margin (%) = (Total Profit / Total Sales) × 100

The total profit displayed in the worksheet was 332.9 and total sales were 3674.6, giving an overall profit margin of approximately 9.06%.

Result: The map is intended to compare profit by region and provide sales details in the tooltip.



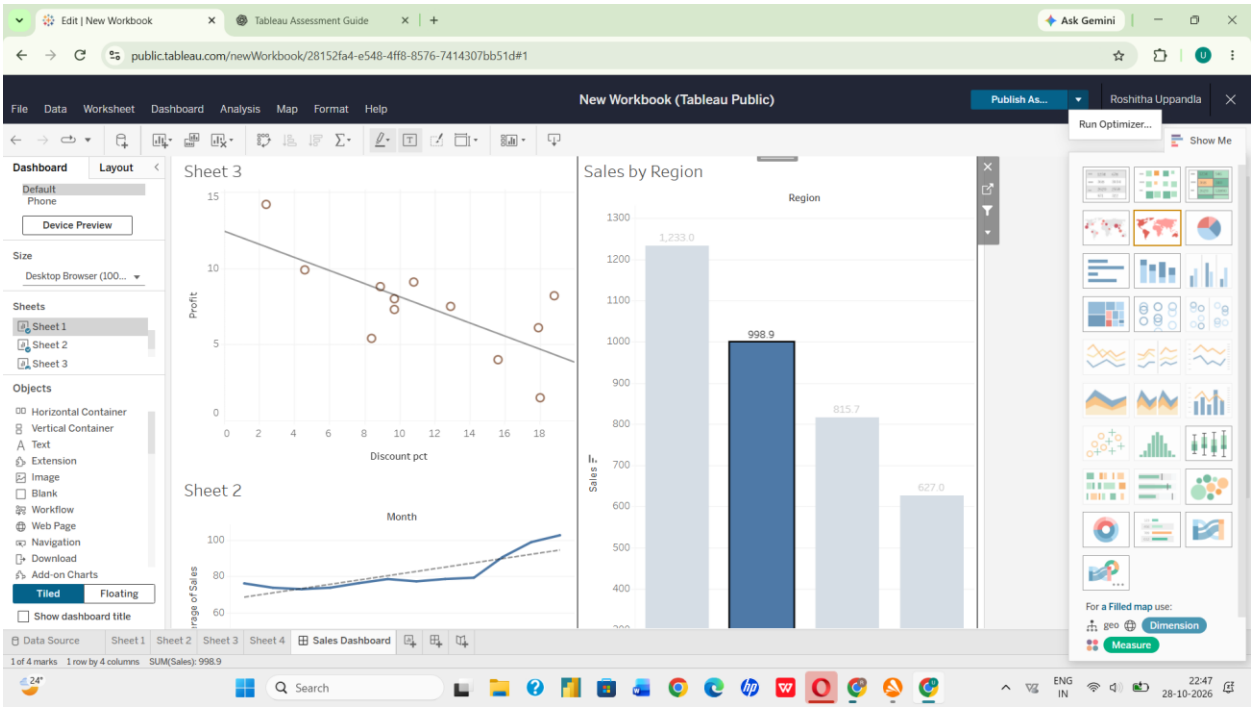
T6: Dashboard

The dashboard brings the individual visualizations together in one place. The bar chart, monthly sales trend and filled map can be added to the same dashboard so that the information can be reviewed together. A Region filter can be applied through the bar chart to make it easier to focus on a particular region.

When a region is selected, the related charts should update or filter according to the dashboard settings. This makes the dashboard more interactive and helps compare the selected region with the overall results.

Result: The dashboard combines the charts and provides a convenient way to explore the data. Add the final dashboard screenshot and the Tableau Public link after publishing.

Tableau Public link: _____



T7: Tableau and R – Two Differences

- Tableau provides a visual drag-and-drop interface for creating charts, while R generally requires code to create visualizations, for example by using ggplot2.
- Tableau makes interactive dashboards and filters straightforward to build. R offers greater flexibility for customized statistical analysis and data processing through programming.

Both tools are useful for data analysis. Tableau is convenient for interactive visual reporting, while R is useful when more control over statistical methods and code is needed.

Conclusion

Through this assessment, I practised connecting a CSV file to Tableau and using the available fields to create visualizations. The bar chart helped compare sales by region, the line chart showed the monthly sales pattern, and the scatter plot helped explore discount and profit. The map and dashboard can provide a regional view and bring the results together. This practical work improved my understanding of data visualization and how charts can be used to explain information.